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# Replicating FTS-Diffusion: Degeneracy of the Released Generation Pipeline

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## Abstract

We revisit FTS-Diffusion (ICLR 2024), a financial time-series generator built from scale-invariant subsequence clustering (SISC), pattern-conditioned diffusion, and learned pattern evolution, and try to reproduce it from the released code. In our runs the released sampling pipeline is near-degenerate: synthetic S&P 500 paths are one repeated motif tiled into an almost perfectly linear price ramp (linear-fit  $R^2 \approx 1.0$ ), independent runs are nearly identical, and synthetic return volatility is about  $20\times$  too small. We trace this to three choices in the released sampling code: a deterministic argmax rollout of the pattern chain, a one-unit decoder hidden state, and a commented-out magnitude normalization. We are explicit about confounds we cannot rule out: no checkpoints are shipped, and our training budget is reduced. Consistent with degeneracy, a continuous-rollout Train-on-Augmentation-Test-on-Real (TATR) setup appears to improve S&P 500 forecasting by up to 85%, yet the same protocol never beats its baseline on GOOG or ZC=F. Augmentation carrying real structure would help on all three assets; a gain limited to the one trending asset points to a scale coincidence. We also do not reproduce the Appendix B.3 SISC toy targets. We present these as observations from one reproduction attempt rather than a verdict on the method.

## 1 Introduction

Deep generative models are attractive for finance because historical market data are scarce, noisy, nonstationary, and not experimentally reproducible. Financial returns also exhibit heavy tails, weak raw-return autocorrelation, volatility clustering, and recurring shapes at irregular time scales [Cont, 2001]. Sequence generators try to preserve temporal structure with adversarial, optimal-transport, and score-based objectives [Yoon et al., 2019, Xu et al., 2020, Tashiro et al., 2021]; financial generators additionally need to respect stylized facts and economically meaningful dependence [Wiese et al., 2020]. FTS-Diffusion [Huang et al., 2024] addresses this by discovering variable-length motifs and generating new trajectories motif by motif.

We reproduce FTS-Diffusion and extend its evaluation; we do not propose a new generator. We retrace the released pipeline and try several generation protocols, asking whether any rollout produces non-degenerate samples. We also test whether the SISC toy-data targets reproduce. The architecture is sensible, but in our hands the released sampling pipeline degenerates: the synthetic series collapses to a near-linear price ramp built from one repeated motif (Section 4). Given that collapse, the apparent S&P 500 gain probably reflects a scale coincidence that does not transfer across assets. We cannot fully separate this from a known confound: the original authors release no trained checkpoints, so we train every module ourselves and at a reduced budget relative to their defaults.

**Contributions.** We make two contributions. (i) A code-level diagnosis of the released sampling pipeline (Section 4): three specific choices that drive the synthetic series to a near-degenerate linear

ramp, plus the underspecifications we had to resolve. (ii) A 30-seed S&P 500 TATR study, with the cross-asset observation that the continuous-rollout gain does not transfer to GOOG or corn futures (ZC=F), which we read as a scale coincidence. We also document a gap in reproducing the Appendix B.3 SISC toy-data targets. Throughout, we separate claims about the released artifacts from what would need the authors’ checkpoints to settle.

## 2 Background: FTS-Diffusion

FTS-Diffusion decomposes financial time-series generation into three modules [Huang et al., 2024]. Scale-invariant subsequence clustering (SISC) segments a univariate series into variable-length motifs using dynamic time warping (DTW) [Sakoe and Chiba, 1978] and DTW barycenter-style centroid updates [Petitjean et al., 2011]. Each segment is summarized by a pattern identity  $p_m$ , a duration scale  $\alpha_m$ , and a magnitude scale  $\beta_m$ . A pattern generation module (PGM) then produces concrete segments from these motif conditions; it pairs a scaling autoencoder (Scaling-AE), which encodes and rescales segment shapes, with a pattern-conditioned diffusion model (PCDM) trained on a DDPM-style denoising objective [Ho et al., 2020].

Long-horizon behavior is controlled by the pattern-evolution module (PEM), which learns

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}), \quad (1)$$

with a classification loss for the next pattern and regression losses for the next duration and magnitude scales. Although useful, this separation makes downstream results sensitive to how the learned state chain is initialized, reset, and divided into synthetic training blocks.

## 3 Replication protocol

**Assets and splits.** Our main datasets are daily Yahoo Finance close-price histories for S&P 500, GOOG, and ZC=F. S&P 500 spans 1980–2020 and yields 10,088 trading days and 2,282 post-SISC segments with  $K = 14$ . GOOG spans 2004–2020 with 3,776 trading days and 733 post-SISC segments with  $K = 11$ . ZC=F spans 2000–2020 with 4,764 trading days and 629 post-SISC segments with  $K = 11$ . We follow a chronological train/test design: motif extraction, fitted generator components, standardizers, and downstream predictors are fit only on their allowed training windows, and final scores are computed on held-out real data.

**Downstream benchmark.** Our main downstream benchmark is TATR: Train on Augmentation, Test on Real. We divide the held-out real slice into an initialization portion and a final real test portion. A one-day-ahead LSTM forecaster is trained on the real initialization portion plus  $Y$  synthetic trading years, where one synthetic year has 252 days and  $Y$  is swept across a fixed grid. We score with mean absolute percentage error (MAPE) on the final real test portion. We use paper-like LSTM settings: one-day ahead, window size 64, hidden size 32, two layers, MAE training loss, Adam with learning rate  $10^{-2}$ , and 100 epochs for the primary S&P 500 runs.

**Synthetic rollout variants.** We compare three ways to produce the synthetic augmentation path, each sharing the same data budget. We use the descriptive name with the code identifier in parentheses.

- **Independent fixed blocks** (authors): independent 252-day blocks, each generated from the same fixed initial state.
- **Continuous chunked** (split): one continuous trajectory from the pattern-evolution chain, cut into 252-day chunks for downstream training.
- **Independent blocks with random init** (random\_init): independent 252-day blocks, each from a randomly drawn initial segment.

These three protocols differ only in how the Markov chain transitions across blocks. Because the pattern-evolution module is the only component that carries long-horizon state across motifs, block connectivity is what they probe.

**Compute.** Our primary 30-seed S&P 500 downstream batch used stored SISC artifacts and PGM/PEM checkpoints and recorded 10,478 seconds on an Apple M4 laptop with 10 CPU cores and 16 GB memory. This time excludes earlier exploratory runs not used for the reported tables.

## 4 Degeneracy of the released generation pipeline

We reproduce FTS-Diffusion end to end from the released code. Before downstream scores, we report what the generator actually produces, because it determines how the later numbers should be read. In our runs the released sampling path is near-degenerate: the synthetic price series is a single repeated motif tiled into an almost perfectly straight line. We identify three code choices behind this, with file references into our working copy (fts-diffusion-ref/), then show the degeneracy on the saved trajectories. We treat these as observations about the released artifacts under our setup rather than proofs about the method; Section 4.4 lists the confounds we cannot rule out.

### 4.1 Three choices in the released sampling code

**(1) Deterministic argmax rollout of the pattern chain.** FTS-Diffusion rolls out the pattern-evolution module (PEM) by taking `argmax` for the next pattern and the next length, and the raw regression mean for the next magnitude, with no sampling (fts-diffusion-ref/models/sampling.py, lines 30, 33, 34):

```
pred_pattern = torch.argmax(probabilities_pattern, dim=1).unsqueeze(0)
pred_length = torch.argmax(probabilities_length, dim=1).unsqueeze(0) + 1_min
pred_magnitude = pred[:, n_patterns+range_length:].float()
```

An autoregressive map  $\text{state}_{m+1} = f(\text{state}_m)$  that is deterministic must, iterated from a fixed initial state, converge to a fixed point or a short cycle. Because the paper does not specify the rollout policy (greedy vs. sampling), this greedy reading of the release is defensible, but it removes the stochasticity the Markov-chain description suggests.

**(2) One-unit decoder hidden state.** Scaling-AE uses an LSTM decoder whose hidden dimension is fed from `pgm_params['sae_hidden_dim']`, set to 1 in the released configuration (model\_params.py; the decoder LSTM is defined in scaling\_autoencoder.py). With a single hidden unit the decoder has almost no capacity, and we observe it collapsing to a fixed output that ignores both the conditioning pattern and the diffusion latent.

**(3) Commented-out magnitude normalization.** A sampling step that would rescale the generated shape to unit range before multiplying by the magnitude  $\beta$  is commented out (sampling.py, lines 53–55), so the emitted segment has magnitude  $\beta \times (\text{whatever range the decoder produced})$  rather than  $\beta$ . In our runs that range is on the order of 0.07, far below the intended scale, which removes the magnitude information  $\beta$  was meant to carry.

### 4.2 What the saved trajectories look like

Figure 1 contrasts a real S&P 500 path with one released-pipeline synthetic path. That synthetic series is visually a straight line; a least-squares linear fit gives  $R^2 \approx 0.99999$ . Three quantitative observations accompany the figure. First, independent runs of a configuration are nearly identical, consistent with a deterministic rollout. Second, the PEM state sequence collapses (Table 1 and Figure 2): on S&P 500 a single pattern is used in 99.6% of segments (a fixed point), while GOOG and ZC=F settle into period-2 cycles. Third, inspecting the trained PGM decoder pattern by pattern, the decoder output is near-constant across all patterns (cross-pattern standard deviation  $< 0.001$ ): an up-trend pattern and a down-trend pattern yield essentially the same segment. Together these tile one tiny upward-creeping segment across the horizon, producing exactly the linear ramp in Figure 1. The same holds on the other assets: GOOG and ZC=F continuous rollouts are also near-linear (linear-fit  $R^2 \approx 0.99999$ ), since their period-2 chains alternate between two near-identical segments.

### 4.3 Underspecifications we had to resolve

Several choices that matter for the result are not pinned down by the paper. Table 2 lists the ones we encountered, separating the three that are load-bearing for the degeneracy above from the compute-driven confounds and the remaining setup choices. We kept the released values wherever they existed, so that the run reflects the artifact as published rather than our preferences. We intend this as an audit aid rather than an accusation: underspecification is common, but here it coincides with the failure modes.

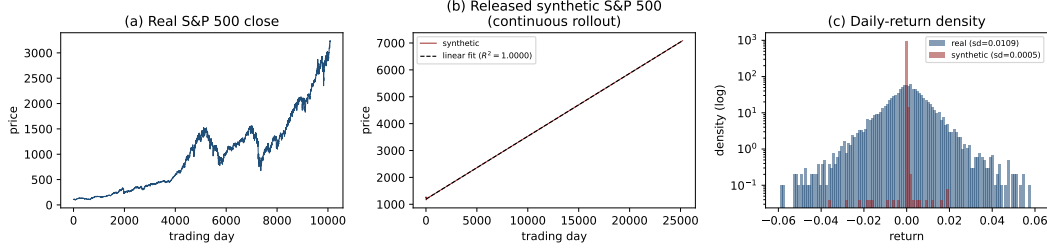


Figure 1: Released-pipeline output on S&P 500. (a) Real close prices show rich structure. (b) One synthetic continuous-rollout path is almost perfectly linear (linear-fit  $R^2 \approx 0.99999$ ). (c) Daily-return density (log scale): synthetic returns concentrate near a single value with standard deviation about  $20\times$  smaller than real returns. We show one representative run; runs are nearly identical.

Table 1: PEM state-sequence collapse under the deterministic rollout, from saved `run_*_states.npy`. “Dominant pattern share” is the fraction of generated segments using the most frequent pattern.

| Asset ( $K$ ) | Segments | Dominant pattern share | Regime         |
|---------------|----------|------------------------|----------------|
| S&P 500 (14)  | 2520     | 99.6%                  | fixed point    |
| GOOG (11)     | 2520     | 50.0%                  | period-2 cycle |
| ZC=F (11)     | 1867     | 50.0%                  | period-2 cycle |

#### 4.4 Confounds and what we cannot rule out

We stress that the evidence above concerns the released artifacts as configured and trained in our hands rather than the method in principle. Two confounds prevent a stronger claim. First, no trained checkpoints are shipped with the release, so every reproduction retrains from the released scripts and we cannot compare against the authors’ own generations. Second, our training budget is reduced relative to the released defaults (PGM 30 vs. 400 epochs, PEM 60 vs. 1000, PCDM 30 vs. 100 steps), a compute-driven choice that a reviewer can reasonably challenge. Three non-exclusive explanations remain that we have not fully separated: undertraining, an unlucky seed, and the one-unit decoder capacity. A full-budget retrain across seeds would falsify or confirm the undertraining hypothesis; we flag it as the natural next experiment and do not assert a method-level conclusion without it. Given how unusual a one-unit decoder hidden state is for a published generative model, it would also be reasonable to ask the authors directly whether that value, and the commented-out magnitude normalization, correspond to the configuration behind the paper’s headline results.

## 5 Replication results

### 5.1 TATR depends on the synthetic-series protocol

Huang et al. report that appending 100 synthetic years reduces one-day-ahead MAPE by 17.9%, 15.3%, and 17.4% on S&P 500, GOOG, and ZC=F, respectively. Our current S&P 500 evidence has three layers. First, the stored result matrices do not support that trend for the documented authors independent-block protocol: the stored S&P 500 authors curve finishes 208.4% worse than its baseline, while the continuous-trajectory curve finishes 71.6% better. Second, a one-seed protocol search found that continuous-rollout variants finish 76.5–85.4% better, whereas independent fixed blocks finish 96.5% worse. Third, Table 3 reports the primary 30-seed batch across the three protocols.

In that batch, the continuous chunked rollout reproduces a strong paper-like reduction in downstream MAPE, while both independent-block protocols worsen it. A single lucky seed does not drive this: across all 30 seeds the final continuous-chunked change stays below baseline (range  $[-89.1\%, -21.2\%]$ ), whereas independent fixed blocks range  $[+17.0\%, +438.3\%]$  and random-init blocks  $[+3.9\%, +263.3\%]$ . Figure 3 shows the same contrast: continuous trajectories cross below the real-only baseline, while independent blocks worsen as augmentation increases.

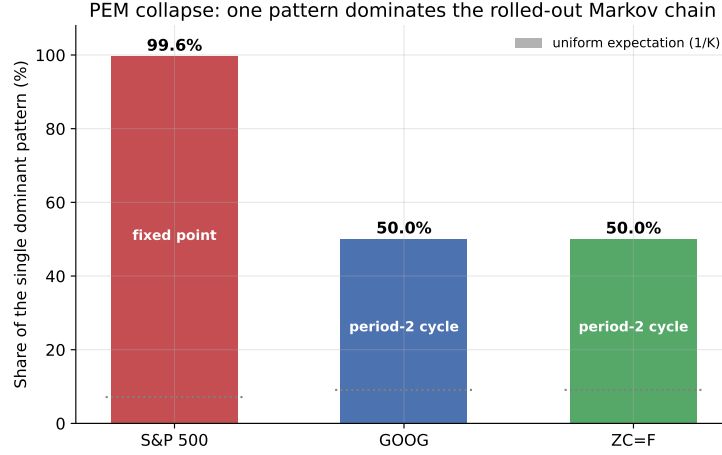


Figure 2: PEM state-sequence collapse under the deterministic rollout. Bars show the share of generated segments that use the single most frequent pattern: 99.6% on S&P 500 (a fixed point) and 50% on GOOG and ZC=F (period-2 cycles). A dashed line marks the uniform expectation  $1/K$  that a non-degenerate chain would approach.

Table 2: Paper-vs-release underspecifications we resolved. “Released” marks a value shipped in the released configuration that we kept; “ours” marks a choice we had to make.

| Design choice                       | In paper? | Value used                          | Note               |
|-------------------------------------|-----------|-------------------------------------|--------------------|
| Scaling-AE decoder hidden dim       | no        | 1 (released)                        | load-bearing       |
| PEM rollout policy                  | no        | argmax (released)                   | load-bearing       |
| Magnitude normalization at sampling | implied   | commented out (released)            | load-bearing       |
| PGM / PEM training epochs           | no        | 30 / 60 (ours; released 400 / 1000) | confound           |
| PCDM diffusion steps                | no        | 30 (ours; released 100)             | confound           |
| SISC $k$ -means seed                | no        | 42 (released)                       | shared seed        |
| GOOG / ZC=F downstream split        | S&P only  | 62.5% train (ours)                  | cross-asset parity |
| Initial segment for generation      | no        | first test segment / random (ours)  | protocol switch    |

Synthetic price levels explain the effect. Continuous trajectories start near 1253 and end near 7085 (mean level 4138); independent blocks start from the same level but repeatedly reset, ending near 1249 (mean 1214). Continuous rollouts thus drift the near-linear ramp through the scale of the upward-trending S&P 500 test period, so percentage error can fall simply because synthetic and real price levels overlap, not because the dynamics are informative. Independent blocks stay near the initial level, the overlap never happens, and error grows. Read with Section 4, this looks like a scale effect rather than useful augmentation. Cross-asset evidence below provides the cleaner test.

Train-on-Mixture-Test-on-Real (TMTR) controls, which train the forecaster on a fixed real-plus-synthetic mixture rather than sweeping the amount of augmentation, make the replication conclusion more conservative. In our TMTR runs, the reference curve reaches a best mean improvement of 26.8% but finishes 359.7% worse; a continuous offset-30 control reaches a best improvement of 50.0% and finishes 17.8% better; a continuous offset-50 control finishes 752.0% worse. So the evidence does not say that synthetic data always helps. A particular continuous TATR price-level rollout can create large apparent gains, while other augmentation and evaluation choices do not.

We make no claim of misconduct; the available generation details do not pin down the downstream result uniquely. Once the generator is seen to be degenerate, though, the simplest reading of the S&P 500 gain is a scale coincidence rather than augmentation that works under the right protocol. Either way, a reproducible benchmark should treat the Markov-chain rollout, initialization, block slicing, scaler fitting, and warm-up windows as first-class parameters, and should not rest a positive conclusion on downstream MAPE for a single trending asset.

Table 3: S&P 500 TATR summary over 30 seeds. Percentages are relative to each protocol’s no-augmentation baseline; lower MAPE is better. “Best change” is the minimum over the augmentation sweep.

| Protocol                           | Baseline | Best change | Final change |
|------------------------------------|----------|-------------|--------------|
| Independent fixed blocks (authors) | 0.0611   | +0.0%       | +208.4%      |
| Continuous chunked (split)         | 0.0620   | −77.6%      | −72.2%       |
| Random-init blocks (random_init)   | 0.0611   | +0.0%       | +68.2%       |

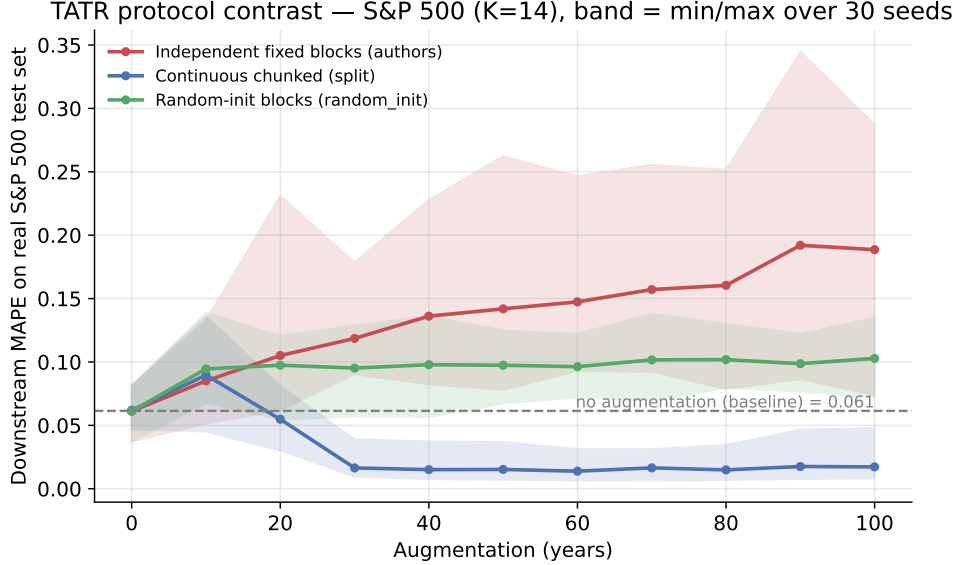


Figure 3: TATR outcome changes sharply with the synthetic-series protocol. Continuous trajectories can look paper-like, while independent blocks worsen with augmentation. A dashed line marks the original S&P 500 100-year TATR target, and shaded bands show min/max across seeds.

GOOG and ZC=F test the coincidence reading most directly. We regenerated the stated real-asset SISC and architecture artifacts for both with  $K = 11$ ,  $l_{\min} = 10$ , and  $l_{\max} = 21$ . A strict author-faithful downstream TATR rerun is blocked by the released S&P-style split logic, which consumes  $252 \times 5 = 1260$  held-out initialization points whereas the GOOG and ZC=F 80/20 held-out windows contain 744 and 963 points, so we adapted the split while keeping the proportions comparable. This changes the absolute MAPE level but not the sign of the augmentation effect, since every percentage is computed within the same split relative to that protocol’s own no-augmentation baseline. Under that within-split comparison the effect is negative on both assets and for both protocol families. Independent fixed blocks (authors) finish 1003% worse on GOOG and 75% worse on ZC=F. Nor does the continuous chunked (split) rollout that appears to help on S&P 500 transfer (Figure 4): its best point over the whole sweep equals the no-augmentation baseline (it never beats it), and it finishes 92% worse on GOOG and 179% worse on ZC=F. If the generator carried useful structure, the protocol would not flip sign across assets; a gain confined to the one strongly trending asset points to a scale coincidence, not augmentation value. We therefore treat GOOG and ZC=F as covered for SISC/architecture setup and real-asset diagnostics rather than author-faithful downstream replications, but the within-split sign does not depend on author-faithfulness.

## 5.2 SISC toy-data replication gap

We also stress-tested SISC on the Appendix B.3 simulated-data setting. Two metrics matter here. Per-unit DTW is the dynamic-time-warping distance between a recovered pattern centroid and its target shape, so lower is better. Jaccard is a segmentation-agreement measure,  $|A \cap B|/|A \cup B|$ , between the recovered set of segment boundaries  $A$  and the ground-truth set  $B$ : it equals 1.0 when the two segmentations coincide and 0 when they share no boundary, and we report a tolerance-2 variant

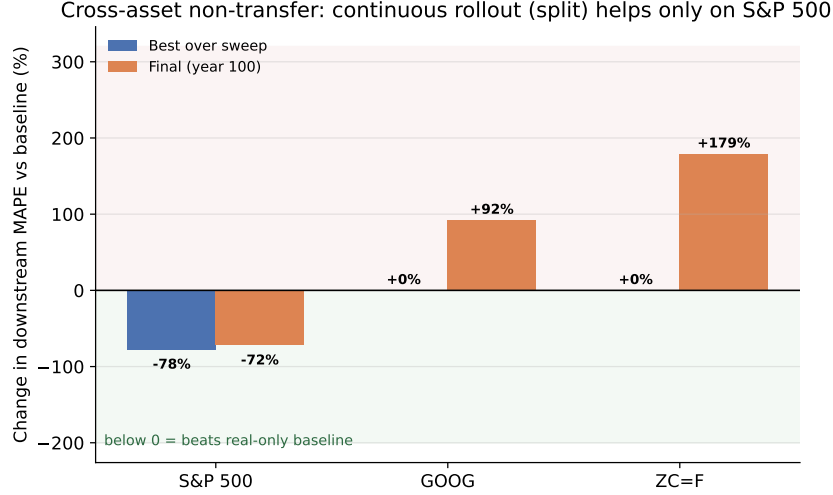


Figure 4: Cross-asset non-transfer of the continuous chunked (split) rollout. Bars show the change in downstream MAPE relative to each asset’s no-augmentation baseline (below zero beats the real-only baseline). The rollout helps only on S&P 500 (best  $-78\%$ , final  $-72\%$ ); on GOOG and ZC=F it never beats baseline and finishes  $+92\%$  and  $+179\%$ . A protocol carrying genuine structure would not flip sign across assets.

that counts a recovered boundary as matched if it falls within two time steps of a true one. Huang et al. report one-pattern per-unit DTW/Jaccard of  $0.009/0.938$  and multi-pattern average per-unit DTW/Jaccard of  $0.01/0.784$ . On our derived one-pattern runs, the best available DTW is  $0.0202$  and the best tolerance-2 boundary-Jaccard proxy is  $0.8693$ . Across 18 successful multi-pattern runs varying seed,  $l_{\max}$ , and maximum iterations, our best average per-unit DTW was  $0.0321$ , and our best author-style interval IoU (intersection over union of recovered and true segment intervals) was  $0.6418$ . Figure 5 summarizes the multi-pattern gap.

### Appendix B.3 Replication Gap

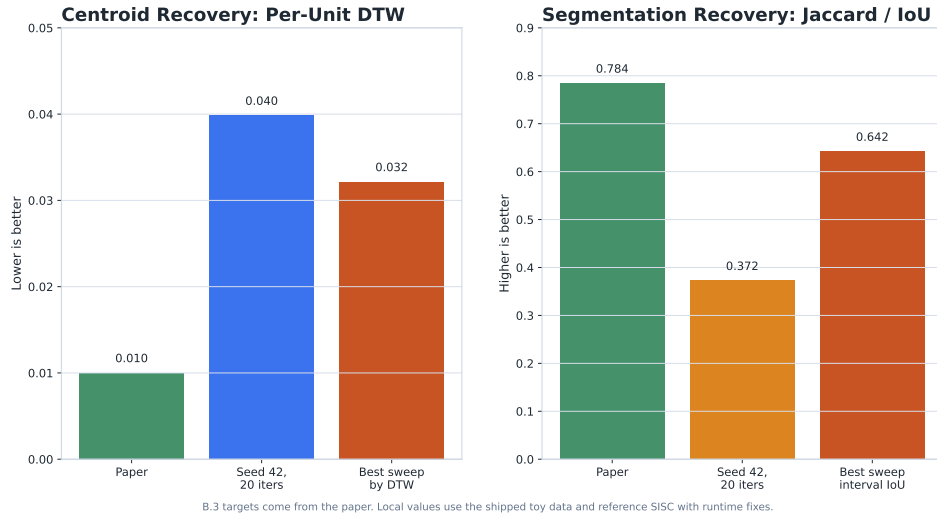


Figure 5: Appendix B.3 SISC replication gap on the toy data. Best replication runs do not reach the reported centroid or segmentation targets.

Increasing SISC iterations from 10 to 20 and changing  $l_{\max}$  from 20 to 21 did not close the gap. Our best strict boundary Jaccard at tolerance 2 was  $0.3608$ , and the best reference-code-style interval IoU (intersection over union of recovered and true segment intervals) was  $0.6418$ . Our best centroid

metric remained about  $3.2\times$  the reported target. We could not reproduce the exact toy benchmark, though this says nothing about the general idea of SISC: the released tree does not expose the exact one-pattern construction, synthetic generator, standard pattern arrays, seed state, or metric implementation used to produce Appendix B.3. More broadly, helper defaults matter: warm-up windows, augmentation scales, plotting conventions, and interval-overlap metrics can all change the apparent result if they go unaudited.

## 6 Limitations and broader impact

Our replication is constrained by the information and compute available for this study. Most importantly, the degeneracy analysis of Section 4 concerns the released artifacts as configured and as trained in our hands: no checkpoints are shipped, and our training budget is reduced relative to the released defaults. We have not run the full-budget, multi-seed retrain that would separate undertraining from an architectural cause, so we deliberately phrase those findings as observations about the release rather than conclusions about the method. Some runs use reference checkpoints rather than a full fresh retraining of all modules. GOOG and ZC=F have shorter histories than S&P 500, so the downstream split used for S&P 500 is not directly portable without changing the question being asked; we mitigate this by comparing within split, relative to each protocol’s own baseline.

Synthetic financial data can be beneficial for stress testing, augmentation, and reproducibility, but it can also create misleading evidence if artificial predictability is mistaken for real market structure. Any release of synthetic paths should clearly label them as generated data, document the training period and generation protocol, and avoid implying that downstream forecasting gains translate into tradable performance.

## 7 Conclusion

FTS-Diffusion has a sensible inductive bias for financial time series: variable-length motifs, pattern-conditioned diffusion, and learned motif evolution. Our reproduction attempt, however, did not yield a working generator from the released artifacts. In our hands the released sampling pipeline collapses to a near-linear ramp built from one repeated motif, and the one downstream setting that appears to improve forecasting does so only on a single strongly trending asset and fails to transfer to GOOG or ZC=F. We read that pattern as a scale coincidence rather than useful augmentation. We do not overstate it: without the authors’ checkpoints and full training budget we cannot yet separate the degeneracy from undertraining, so we report empirical observations about the release, not a verdict on the method. One methodological point holds regardless: realistic-looking paths and a favorable downstream MAPE on a trending series are weak evidence on their own. Future synthetic-finance benchmarks should report exact rollout protocols and cross-asset transfer, rather than resting positive conclusions on a single asset.

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